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EDITORIAL

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Introduction

Artificial Intelligence (AI) in healthcare, initially described in 1950, took a leap in the early 2000s with the advent of deep learning, aiming to emulate human cognitive functions and fuelled by the growing availability of healthcare data[1]. Common AI tools encompass machine learning for data categorization, natural language processing, predictive analytics, and speech recognition technology to enhance provider–patient dialogue[1,2]. These instruments empower healthcare providers to predict future health paths, recommend treatments, and discover new genetic connections[1,2].

However, economic challenges related to AI in healthcare can be formidable, potentially affecting investment choices[3]. Health Economic Evaluations (HEEs) offer critical insights to gauge the financial implications of AI integration. Regrettably, many current HEEs have methodological limitations, necessitating thorough evaluations to guide well-considered decisions[4]. Obstacles include external factors related to the healthcare system, the ability to manage strategic change, and the evolution of healthcare professions and practices – all of which could entail substantial costs[5]. Financial constraints, significant initial expenses, and rigorous economic assessments are among the chief barriers hindering the progression of AI in healthcare.

Conversely, AI promises considerable cost savings. Estimates suggest potential savings between $200 billion to $360 billion solely within the US when applied across various healthcare sectors[6].

The long-term economic implications of AI in healthcare warrant crucial discussions, yet these topics remain underrepresented in the academic works. This editorial aims to bridge this gap by examining the current studies in the literature. Grasping the long-term economic impact of AI in healthcare can furnish stakeholders with priceless knowledge, paving the way for more informed strategic planning and judicious investment choices. Identifying areas where AI can maximize cost-effectiveness and elevate patient care will enable more precise resource allocation. Moreover, delving into its extended financial consequences can unveil potential returns on AI healthcare investments, presenting a more transparent financial overview for stakeholders contemplating such ventures.

Discussion

This editorial presents a comprehensive exploration of the long-term economic implications of AI in healthcare, systematically and unravelling these effects from various perspectives. The analysis is organized around eight key dimensions: Cost Savings, Cost Avoidance, Long-term Health Outcomes, Innovation and Research, and External Economic Factors and Strategic Transformation, with each dimension further delineated into subcategories. Our discussion spans a spectrum of economic considerations tied to AI in healthcare, ranging from direct cost savings and revenue enhancement to strategic and external factors. Through this methodical evaluation, using a comprehensive multi-criteria framework, we aim to furnish a holistic view of how AI might economically shape healthcare in the future.

Cost savings

AI-enabled healthcare promises significant savings to healthcare systems and patients, stemming from enhanced diagnostic, operational, and treatment efficiencies, as well as preventive care facilitated by AI. These savings, primarily benefiting healthcare providers and patients, suggest a remarkable economic return over time, with enduring positive implications for the entire healthcare ecosystem. One study employed a deterministic model to evaluate various testing strategies for first-line metastatic colorectal carcinoma treatment. It found that an AI approach complemented by high-specificity panels resulted in substantial savings – $400 million or a 12.9% reduction compared to using next-generation sequencing alone[7]. Moreover, projections indicate that AI integration might lead to annual healthcare savings ranging between $200 billion and $360 billion, underscoring its enduring fiscal benefits[8]. The transformative potential of AI in healthcare is further exemplified by its ability to tailor patient care. This is highlighted by a 25% reduction in hospital stays and a staggering 91% decrease in nursing facility discharges, which translates to both economic savings and enhanced patient outcomes[9]. Preventive healthcare augmented by AI has a profound positive impact on medical expenses, promoting early disease identification and timely interventions, which in turn diminish the need for hospitalizations and expensive procedures. For instance, innovative AI technologies can pinpoint diseases early on, ensuring swift and effective treatments that result in fewer hospital stays and overall reduced costs. An analysis on the cost-effectiveness of early Parkinson's Disease (PD) detection *via* avant-garde technology posited that such early diagnosis not only elevates patient quality-of-life but also curtails treatment expenditures[10].

However, the current research exhibits limitations, including potential biases, limited sample sizes, and a predominant emphasis on immediate outcomes, often overlooking long-term economic ramifications. Additionally, variations across healthcare systems can yield divergent outcomes, complicating broad generalizations.

Table 1. Potential long-term economic impacts of AI in healthcare

Categories	Subcategories	Example
Cost Savings	Diagnostic efficiency	AI-powered imaging tools that instantly analyse and interpret X-rays, MRIs, or CT scans, reducing the need for multiple radiologists to review images and thus reducing costs.
	Operational efficiency	AI-driven appointment scheduling systems that optimize patient flow and reduce waiting times. Such systems can predict no-shows and overbook accordingly, resulting in fewer wasted slots and maximizing revenue for clinics.
	Treatment efficiency	Personalized treatment plans generated by AI, based on a patient's genetics, medical history, and other relevant data. This can lead to more effective treatments that have a higher likelihood of success, reducing the need for prolonged or multiple treatments.
	Preventive care	Wearable devices integrated with AI algorithms that monitor vital signs and predict potential health issues before they become serious.
Cost avoidance	Compliance and Litigation	AI-powered systems that ensure medical practices and procedures strictly adhere to established guidelines, decreasing the chances of non-compliance. This not only avoids financial penalties but also reduces the likelihood of malpractice incidents.
	Readmission Penalties	AI-driven clinical pathways, such as Clinical Decision Support Systems (CDSSs), can be implemented to monitor and manage chronic ailments.
	Waste Reduction	AI systems can be utilized to analyse vast amounts of patient and procedural data to recommend or de-prioritize certain tests and procedures based on their anticipated value for a specific patient.
	Resource Optimization	AI can aid in refining resource utilization, ensuring that equipment and human resources are used to their utmost potential and avoiding unnecessary expenditures.
Long-term Health Outcomes	Quality of Life	AI-powered remote monitoring tools allow patients with chronic conditions like diabetes to continuously monitor their glucose levels without frequent hospital visits.
	Population Health Management	A decrease in healthcare expenses due to proactive management, and a reduction in economic downturns which often accompany large-scale disease outbreaks because of decreased productivity and increased healthcare costs.
	Longevity	AI can analyse molecular structures and predict their potential impact on human longevity, leading to the faster discovery of anti-aging compounds. The economic impact of increased longevity means a longer productive life for citizens, potentially extending their time in the workforce and their contributions to the economy.
Innovation and Research	Technological Breakthroughs	AI-driven laboratories that can automate and expedite drug discovery processes, leading to faster introduction of new therapeutic agents., which leads to cost savings
	Spin-off Innovations	AI innovations initially designed for healthcare, such as wearable health monitors, can find applications in other industries, like sports or fitness.
	Economic Ripple Effects	The development and commercialization of AI-driven healthcare tools can stimulate local economies by creating jobs in tech, manufacturing, and distribution sectors.
Market Competition Dynamics	Differentiated Services	Hospitals using AI-driven diagnostic tools can offer more precise treatment plans, differentiating themselves from competitors.
	Enhanced Patient Choice	AI can provide personalized treatment options, giving patients the ability to choose the best fit for their needs.
	Cost-effective Healthcare Models	AI-driven telemedicine platforms can provide expert consultations at a fraction of traditional costs, making healthcare more affordable.

Cost avoidance

The application of AI in healthcare serves as a cornerstone for cost avoidance, primarily aiding healthcare providers and insurance companies, significantly mitigating financial burdens. This cost avoidance, particularly beneficial to healthcare providers and insurance agencies, reflects a reduction in unnecessary expenses and legal costs, different from direct savings achieved through operational efficiencies. Artificial Intelligence aids in compliance measures and reduces malpractice incidents, thus dramatically lowering litigation expenses. Tools such as Google DeepMind exhibit exceptional accuracy in disease diagnosis, thereby reducing diagnostic errors that frequently lead to malpractice lawsuits. AI also plays a pivotal role in reducing readmission penalties; research illustrates that AI-driven clinical pathways, including Clinical Decision Support Systems (CDSSs) for chronic ailments like COPD, can markedly decrease readmission rates, thus lessening penalties and related expenses[11,12]. Another prime advantage of AI is waste reduction; it fine-tunes resource distribution and curbs unnecessary tests and procedures, leading to considerable cost avoidance. AI complements systems like the Lean/Toyota Production System (TPS), renowned for its efficacy in eradicating waste and refining resource utilization[13]. Through these channels, AI not only alleviates financial pressures but also steers healthcare toward more streamlined, compliant, and resource-wise practices, highlighting its invaluable position in contemporary healthcare cost management.

Long-term health outcomes

The potential of AI in healthcare offers immense promise for enhancing patient quality-of-life. AI's capability to analyze patient data rapidly and with greater accuracy than humans promises precise diagnoses, individualized treatment plans, swifter recovery periods, and fewer complications. Such advancements also bring economic benefits by minimizing resources squandered on misdiagnoses or ineffective treatments.

In population health management, AI stands out by both forecasting health trajectories and optimizing community health results, simultaneously easing strains on healthcare infrastructures. BlueDot, a Canadian firm, uses AI to assess varied data points, including news headlines, airline ticket sales trends, and animal disease outbreaks, aiming to predict

potential outbreaks[14]. Notably, this company alerted users to COVID-19 risks even before the WHO made its public announcement.

Moreover, AI is poised to be instrumental in our pursuit of extended life spans. It can bolster endeavors to prolong healthy life expectancies, allowing people to stay economically active for longer durations. Innovations powered by AI in areas like genetics, drug discovery, and personalized medicine harbor immense potential to devise treatments that can lengthen the healthful stages of human existence. Atomwise, a company focused on AI-driven drug discovery, has pioneered technology aimed at predicting how effective specific molecules might be against particular diseases[15]. Ideally, this approach could diminish both the time and costs tied to conventional drug discovery, potentially hastening the availability of groundbreaking treatments.

AI has become a vital tool in drug discovery, offering a promising approach to rapidly identify new candidates and accelerate the drug development process. Previously, drug development was fraught with uncertainty, with only 10% of candidates advancing beyond the first stage of clinical trials and a timeline of 10–15 years for commercialization[16]. AI presents a solution that could reduce these timeframes and costs significantly[16]. Economically, AI has been crucial in drug discovery. AI-first biotech companies have experienced significant growth in their drug pipelines through this method. Currently, over 150 small molecule drugs are in discovery and 15 clinical trials using AI technologies are in progress, with this number increasing by nearly 40% annually[17]. AI offers value by accessing new biology, enhancing chemistry, increasing success rates, speeding up discovery processes, and reducing discovery costs (BCG, n.d.). Investment in AI-enabled drug discovery has surged, with third-party investment exceeding $5.2 billion by the end of 2021[17].

Other areas

AI's involvement in innovation and research, especially within healthcare, is multifaceted. It has the potential to drive technological breakthroughs, resulting in spin-off innovations that give rise to new healthcare products and services. The ripple effects of these innovations might transcend medical contexts, influencing broader economic landscapes.

The prospective impact of AI on market competition dynamics within healthcare is compelling. Its ability to create differentiated services can both enhance patient choice and pave the way for cost-effective healthcare models. While AI has the potential to nudge healthcare organizations toward patient-centric innovations, emerging regulatory challenges present obstacles. These challenges may stymie innovation or stifle market competition. However, well-designed regulatory frameworks can balance market risks and foster sustainable economic models in healthcare.

Considering strategic positioning and digital transformation, AI's long-term economic contributions might be profound. By leveraging AI, healthcare entities can refine their strategic stance while simultaneously undertaking digital transformations that challenge traditional healthcare service delivery paradigms. The potential for streamlined operations, elevated patient care, and novel revenue channels underscores AI's revolutionary capacity, extending the economic conversation beyond the confines of the healthcare domain.

In economic discussions centered on healthcare, AI emerges as both a primary actor and a facilitator. The interplay between AI, innovation, market competition, and regulatory frameworks weaves a complex story, emphasizing AI's long-term economic implications in healthcare. This narrative is embedded within broader dialogues on technology, economics, and healthcare delivery, spotlighting the intricate opportunities and challenges AI presents within healthcare environments. This layered narrative underscores the necessity for further exploration of the vast potentials AI holds for the healthcare industry at large.

Conclusion

AI offers the potential for significant cost reductions through improved diagnostic accuracy and optimized resource allocation, while also elevating patient care quality *via* personalized treatment models. In this editorial, we delved into the potential savings attributable to AI solutions. However, regulatory challenges and variations in healthcare systems across regions may hinder the seamless integration of AI and its economic benefits. Additionally, the strategic transformation spurred by AI indicates that its economic impact extends beyond just healthcare. AI's intricate interplay with market dynamics and regulatory frameworks gives rise to a nuanced economic discussion that warrants further exploration. Such an in-depth analysis will guide stakeholders in strategic planning, investment decisions, and provide a clearer understanding of the economic opportunities and challenges inherent in an AI-driven healthcare landscape.

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Author contributions

AZA developed the study design, performed data extraction, manuscript drafting, and reviewing.

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References

[1] Kaul V, Enslin S, Gross SA. History of artificial intelligence in medicine. Gastrointest Endosc. 2020;92(4):807–812. doi: 10.1016/j.gie.2020.06.040.

[2] Jiang F, Jiang Y, Zhi H, et al. Artificial intelligence in healthcare: past, present and future. Stroke Vasc Neurol. 2017;2(4):230–243. doi: 10.1136/svn-2017-000101.

[3] Wolff J, Pauling J, Keck A, et al. The economic impact of artificial intelligence in health care: systematic review. J Med Internet Res. 2020;22(2):e16866. doi: 10.2196/16866.

[4] Voets MM, Veltman J, Slump CH, et al. Systematic review of health economic evaluations focused on artificial intelligence in healthcare: the tortoise and the cheetah. Value Health. 2022; 25(3):340–349. doi: 10.1016/j.jval.2021.11.1362.

[5] Petersson L, Larsson I, Nygren JM, et al. Challenges to implementing artificial intelligence in healthcare: a qualitative interview study with healthcare leaders in Sweden. BMC Health Serv Res. 2022;22(1):850. doi: 10.1186/s12913-022-08215-8.

[6] McKendrick J. Healthcare may be the ultimate proving ground for artificial intelligence. Forbes. 2023. https://www.forbes.com/sites/joemckendrick/2023/02/22/healthcare-may-be-the-ultimate-proving-ground-for-artificial-intelligence/?sh=39cea43c2b55

[7] Kacew AJ, Strohbehn GW, Saulsberry L, et al. Artificial intelligence can cut costs while maintaining accuracy in colorectal cancer genotyping. Front Oncol. 2021;11:630953. doi: 10.3389/fonc.2021.630953.

[8] Pifer R. Artificial intelligence could save healthcare industry $360B a year. 2023. https://www.healthcaredive.com/news/artificial-intelligence-healthcare-savings-harvard-mckinsey-report/641163/#:~:text=In the new paper%2C researchers estimate that broader,next five years%2C without sacrificing quality or access.%23:~:text=In the new pap

[9] Lagasse J. How artificial intelligence can be used to reduce costs and improve outcomes in total joint replacement surgery. Healthcare Finance. 2023. https://www.healthcarefinancenews.com/news/how-artificial-intelligence-can-be-used-reduce-costs-and-improve-outcomes-total-joint#:~:text= Use of AI led to, experimenting with to reduce expenses

[10] Munoz DA, Kilinc MS, Nembhard HB, et al. Evaluating the cost-effectiveness of an early detection of parkinson's disease through innovative technology. Eng Econ. 2017;62(2):180–196. doi: 10.1080/0013791X.2017.1294718.

[11] Romero-Brufau S, Wyatt KD, Boyum P, et al. Implementation of artificial Intelligence-Based clinical decision support to reduce hospital readmissions at a regional hospital. Appl Clin Inform. 2020;11(4):570–577. doi: 10.1055/s-0040-1715827.

[12] Wang L, Li G, Ezeana CF, et al. An AI-driven clinical care pathway to reduce 30-day readmission for chronic obstructive pulmonary disease (COPD) patients. Sci Rep. 2022;12(1):20633. doi: 10.1038/s41598-022-22434-3.

[13] AHRQ. Reducing waste and inefficiency in health care through lean process redesign: literature review. 2018. https://www.ahrq.gov/research/findings/final-reports/leanprocess/leanprocess.html#:~:text=Lean/Toyota

[14] Borda A, Molnar A, Neesham C, et al. Ethical issues in AI-enabled disease surveillance: perspectives from global health. Appl. Sci. 2022;12(8):3890. doi: 10.3390/app12083890.

[15] Atomwise. Artificial intelligence for drug discovery. 2023. https://www.atomwise.com/

[16] Yu KH, Beam AL, Kohane IS. Artificial intelligence in healthcare. Nat Biomed Eng. 2018;2(10):719–731. doi: 10.1038/S41551-018-0305-Z.

[17] Chun M. How artificial intelligence is revolutionizing drug discovery. Harvard Law School. 2023. https://blog.petrieflom.law.harvard.edu/2023/03/20/how-artificial-intelligence-is-revolutionizing-drug-discovery/

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